

Machine Learning

Why was Machine Learning Created?

- Before Machine Learning, software relied entirely on manually written rules created by programmers. These rules worked well for simple and predictable tasks but became difficult to manage when dealing with complex real-world problems.

Identifying Cats in Images

Traditional Programming:

- IF Animal has whiskers
- AND Animal has pointed ears
- AND Animal has four legs
- THEN Cat

Problems:

- Every cat looks different.
- Some dogs have similar features.
- Cats can appear in different positions, sizes, and lighting conditions.
- Writing rules for every possible situation is nearly impossible.
- The number of rules becomes very large and difficult to maintain.
- Traditional programs often fail when they encounter new or unexpected examples.

Traditional Programming vs Machine Learning

Traditional Programming

- In Traditional Programming, humans write all the rules
- Input Data + Rules -> Output

Example: Suppose you want to identify whether a person is eligible for a loan.

you write Rules:

- if salary > 50000 and credit_score > 700
 - approve_loan()
- else:
 - reject_loan()

The computer only follows your rules.

Machine Learning

in ML, The Computer Learns the Rules from Historical Data.

Input Data + Correct Answers -> ML Model

Evolution of Machine Learning

Stage 1: Data Collection

- Organizations collect large amount of data.

Examples:

- Customer Purchases
- Student Marks
- Medical Records
- Bank Transactions

Stage 2: Learning

- The Machine Learning analyzes historical Data.

Stage 3: Prediction

Once trained, the model can make predictions on new unseen data.

Example:

Past Data:

Hours Studied	Marks
2	40
4	55
6	70

New Input:

Hours = 5

Prediction:

Marks ≈ 62

Common Machine Learning Applications

| Applications | Example | |-----|-----| |
Recommendation Systems | Netflix, Amazon | Fraud Detection |
Credit Card Fraud | | Medical Diagnosis | Disease Prediction | |

Image Recognition | Face Unlock | | Speech Recognition | Siri,
Alexa | | Spam Detection | Gmail |

Types of Machine Learning

Machine Learning Algorithms are mainly classified into four types based on the availability of labeled data and the learning process.

- Supervised Learning
- Unsupervised Learning
- Semi-Supervised Learning
- Reinforcement Learning

1. Supervised Learning

Supervised Learning is a type of Machine Learning where the model is trained using Labeled Data.

Labeled Data=Input + Correct Output

The Algorithm learns the relationship between input variables (features) and output variables (target).

Example:

Student Dataset

Study Hours	Marks
2	40
4	60
6	80

The Model Learns:

- More Study Hours -> Higher Marks

Now if a Student Studies 5 Hours:

Predicted Marks≈70

Real-Life Applications

- House Price Prediction
- Loan Approval
- Disease Prediction
- Student Performance Prediction
- Spam Email Detection

Types of Supervised Learning

1. Regression

Predicts Continuous Numerical Values.

Examples:

- House Prediction
- Salary
- Temperature

Output=Number

2.Classification

Predicts Categories.

Examples:

- Spam/Not Spam
- Pass/Fail
- Male/Female

Output=Category

Advantages:

- High Accuracy when labeled data is available.
- Easy to evaluate performance.
- widely used in industry.

Disadvantages:

- Requires large labelled datasets.
- Labelling data is time-consuming.

2. Unsupervised Learning

- Unsupervised Learning is a type of Machine Learning where the model learns from Unlabelled Data.
- Only inputs are available, no correct answers are provided.

Example:

- Customer Dataset:

Customer	Age	Income
A	25	30000
B	30	45000

Possible Output:

- Group 1 -> Young Customers
- Group 2 -> High-Income Customers

Main Goal

- Find hidden structures and patterns in data.

Applications:

- Customer Segmentation
- Market Basket Analysis
- Fraud Detection
- Recommendation Systems
- Social Network Analysis

Types of Unsupervised Learning

A.Clustering

- Grouping Similar Observations.

Example:

- Customers with similar buying habits

Algorithm:

- K-Means Clustering

B.Association

- Finding Relationship Between Items.

Example:

- People Who buy Milk also buy bread.
- used in supermarkets.

Advantages

- No Labeled Data Required.
- Helps Discover Hidden Patterns

Disadvantages:

- Harder to Evaluate
- Results may be difficult to interpret.

3. Semi-Supervised Learning

- Semi-Supervised Learning Combines Both:
- Small Amount of Labeled Data + Large Amount of Unlabelled Data.

Why Use it?

- Labeling Data is expensive and Time Consuming.

Example:

- Hospital Has:
 - 1,000 X-ray images labeled by doctors (e.g., Normal or Pneumonia)
 - 50,000 X-ray images without labels

Instead of Labelling all images, the model learns from both labelled and unlabeled data.

Example:

Suppose we have: | Image | Label | |-----|-----| | Cat Image | Cat | | Dog Image | Dog | |
Thousands of Images | No Label |

Applications:

- Medical Image Analysis
- Speech Recognition
- Face Recognition
- Text Classification

Advantages:

- Requires fewer labeled records
- Better Performance than purely unsupervised learning
- Reduces labeling cost.

Disadvantages:

- More Complex to Implement
- Incorrect Labels can affect Performance.

4.Reinforcement Learning

- Reinforcement Learning (RL) is a Learning Technique where an **agent learns by interacting with an environment and receiving rewards or penalties.**

Basic Concepts:

Agent -> Action -> Environment -> Reward/Penalty

- The goal is to Maximize Rewards.

Example: Self-Driving Car

Actions:

- Turn Left
- Turn Right
- Brake
- Accelerate

Applications:

- Self-Driving Cars
- Robotics
- Game Playing AI
- Route Optimization
- Industrial Automation

Advantages:

- Learns from Experience
- Handles dynamic environments.
- Can achieve very high performance.

Disadvantages:

- Requires many training iterations.
- Computationally Expensive.
- Complex to design reward systems.

Comparison Table

Feature	Supervised	Unsupervised	Semi-Supervised	Reinforcement
Labeled Data	Yes	No	Partial	No
Target Variable	Available	Not Available	Partial	Reward Signal
Goal	Prediction	Pattern Discovery	Better Learning	Maximize Reward
Example	House Price Prediction	Customer Segmentation	Medical Imaging	Self-Driving Cars
Common Algorithms	Linear Regression, Decision Tree	K-Means, PCA	Label Propagation	Q-Learning, Deep Q Networks

Machine Learning Workflow

Complete ML Workflow

1. Problem Definition
2. Data Collection
3. Data Preprocessing
4. Feature Selection/Engineering
5. Train-Test Split
6. Model Training
7. Model Evaluation
8. Hyperparameter Tuning
9. Model Development
10. Monitoring and Maintenance

3. Problem Definition

- Real - World data is rarely Clean.
- Data Preprocessing prepares raw data for machine learning

Common Tasks:

1. Handle Missing Values

2. Remove Duplicates
3. Handle Outliers Methods:
 - IQR
 - Z-Score
 - Box Plot
4. Encode Categorical Data
5. Scale Numerical Features Methods:
 - StandardScaler
 - MinMaxScaler
 - RobustScaler

4. Feature Selection and Feature Engineering

What is a Feature?

- A Feature is an Input variable used by the model.

Example:

Predict Student Marks:

Available Features:

- Name
- Roll-No
- Feature Engineering

Creating New

5. Train-Test Split

- We should not train and test on the same data.

Why?

- The model may simply memorize the data.

Standard Split

- Training Data = 80%
- Testing Data = 20%

Example:

1000 Records

- 100 -> Training
- 200 -> Testing

6. Model Training

Training is the process where the algorithm learns patterns from data.

Example:

Student Dataset

Study Hours	Marks
2	40
4	60
6	80

7. Model Evaluation

- After Training, we test the model using unseen data.

Goal:

- Determine how well the model Performs.

Regression Metrics: MAE

- Mean Absolute Error
- Measures average prediction error.

RMSE

R² Score

Classification Metrics

Accuracy

- Correct Predictions
- Total Predictions

Precision

Recall

8. Hyperparameter Tuning

- After Evaluation, we try to improve performance.
- This Process is called Tuning.

Hyperparameter

- A Setting chosen before tuning.

Examples: Decision Tree

- max_depth
- min_samples_split

Random Forest

- n_estimators

- max_depth

Why Tune?

- Different settings produce different results.

Example: Decision Tree

*max_depth=2 Accuracy=75%

max_depth=5 Accuracy=88%

9. Model Deployment

- After achieving good performance, the model is deployed.
- Deployment means making the model available for real users.

Example:

- House Price Model

Input: *Area=1200 sq.ft Bedrooms=3

Output:

- Price=65 Lakhs

Real-Life Examples

1. Netflix Recommendation Engine

10. Monitoring & Maintenance

- Deployment is not the end.
- models must be monitored continuously.

Example:

- Customer behavior in 2023 may differ from 2026
- This is Called:

Data Drift:

- When the input data patterns change. **Concept Drift:**
- When relationships between inputs and outputs change.

Actions

- Retrain model regularly.
- Collect new data
- Update features
- Re-evaluate performance

Concept	Underfitting	Overfitting
Definition	Model too Simple	Model too Complex
Training Accuracy	Low	Very High
Testing Accuracy	Low	Low
Bias	High	Low
Variance	Low	High
Learning Curves	Curves Close and Low	Large gap between curves

Overfitting & Underfitting

Overfitting

Overfitting occurs when a machine learning model learns the training data too closely, including its noise and random variations. As a result, the model performs very well on the training data but poorly on unseen test data.

Characteristics

- Very high training accuracy
- Low test accuracy
- Low bias
- High variance
- Poor generalization

Detection Using Learning Curves

- Training score remains very high.
- Validation score is much lower.
- A large gap exists between the training and validation curves.

Illustration



Underfitting

Underfitting occurs when a machine learning model is too simple to learn the underlying patterns in the data. Consequently, it performs poorly on both training and test datasets.

Characteristics

- Low training accuracy

- Low test accuracy
- High bias
- Low variance
- Poor learning of data patterns

Detection Using Learning Curves

- Both training and validation scores are low.
- The curves remain close to each other.
- Performance does not improve significantly.

Illustration



Bias vs Variance Tradeoff

The **Bias-Variance Tradeoff** is the balance between a model's ability to learn patterns from training data (**bias**) and its ability to generalize to unseen data (**variance**).

The objective is to find a model complexity that minimizes both bias and variance, resulting in the best predictive performance.

Bias

Definition

Bias is the error introduced when a model makes overly simplistic assumptions about the data.

High Bias

- Model is too simple.
- Important patterns are missed.
- Causes underfitting.

Example

Using **Linear Regression** to model a highly nonlinear relationship.

| **High Bias** → **Underfitting**

Variance

Definition

Variance is the error caused when a model becomes too sensitive to the training data and learns noise instead of general patterns.

High Variance

- Model is too complex.
- Memorizes training data.
- Causes overfitting.

Example

A very deep **Decision Tree** fitting every training sample perfectly.

| **High Variance** → **Overfitting**

The Sweet Spot

Definition

The **Sweet Spot** is the optimal balance between bias and variance where the model captures meaningful patterns from the data without memorizing noise.

Characteristics

- Low training error
 - Low validation error
 - Good generalization
 - Best predictive performance
-

Subs close and low |

| **Overfitting** | Model too complex | Very High | Low | Low | High | Large gap between curves |
